# Marginal-Preserving Support Interventions in Synthetic Transformer-Block Distillation

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20 October 2026

**Abstract**

A Transformer hidden vector can already encode substantial context from earlier layers. How much does its cross-position arrangement matter when a compressed block learns from synthetic hidden sequences? We study this question with Q_pos, a support intervention that independently permutes donor rows at each position while preserving the exact finite empirical multiset of complete hidden vectors. Students learn frozen block residual updates from Native or Q_pos synthetic inputs, with fresh Teacher labels and a fixed paired training procedure. Three separately registered campaigns reveal configuration-dependent effects. At GPT-2 block 5, Q_pos produces material losses on both value and history-response fidelity: mean normalized-error penalties are 0.063807 and 0.029738. GPT-2 block 9 meets its registered 5%-of-Native-recovery preservation rule, and Pythia-160M block 5 meets qualified preservation under absolute margins. Exploratory cross-support evaluation locates the GPT-2 block-5 deficit on independent Native synthetic inputs as well: Native- and Q_pos-trained Students score 0.102917 and 0.156757 on that common support, while their ranking reverses on Q_pos support. The study establishes a controlled empirical failure of marginal sufficiency for one fixed learning configuration, alongside preservation cases, and characterizes the resulting support-conditioned approximations. It shows why evaluating learned operators on common supports is essential to understanding synthetic distillation.

## 1 Introduction

Transformer hidden states carry both local information and context accumulated through preceding layers. A complete vector at one position can therefore contain information about the sequence from which it arose. When such vectors are used to train a compressed downstream block, does preserving every position's empirical vector collection preserve the learning problem sufficiently for transfer, or does their arrangement across positions still matter?

This question has a practical consequence for synthetic block distillation. Hidden states generated by a frozen prefix provide training inputs for approximating an internal operator. Their usefulness may depend on information inside individual vectors, on the grouping of vectors into sequences, and on how a restricted learner uses both. Although marginals do not determine a joint distribution, that mathematical fact does not determine the size of a learning effect in a particular Transformer configuration. The effect requires measurement.

We make this measurement using an exact finite intervention. Native hidden banks are generated by passing iid random token-ID sequences through a frozen Teacher prefix. Q_pos independently permutes donor rows at each sequence position, preserving the entire multiset of hidden vectors in each column. Each vector retains its upstream contextual information; its neighboring vectors generally come from different donors. The Teacher then labels the recombined sequences directly. Paired Students share initialization and a fixed learning budget, so the

experiment measures the consequence of changing this organization within a defined learning procedure.

The marginal equality is between the Native and Q_pos **synthetic banks**. Real-language-induced target states provide a separate transfer test. This separation lets us ask whether two synthetic supports with identical position-wise vector collections produce comparably transferable approximations of the same frozen block.

The answer varies across configurations. ID2, at GPT-2 Small block 5, shows material losses in both registered endpoints. XR1, at GPT-2 block 9, retains at least 95% of Native's recovered improvement under its relative criterion. CMR1, at Pythia-160M block 5, achieves qualified preservation under absolute margins. These outcomes identify both a substantive failure case and settings in which recombined supports remain adequate for the registered tolerance.

We further characterize the learned approximations through exploratory cross-support evaluation. Both terminal Students are evaluated on both independent synthetic development banks. In ID2, the Q_pos-trained Student is worse on Native support in all 48 pairs, yet has lower mean error on Q_pos support. Its deficit is thus already visible within the synthetic setting. Own-support fit alone would conceal this reversal of Student ordering.

The study contributes:

1. **Controlled support intervention:** an exact finite position-vector marginal control in Transformer-block distillation, with one formal material-loss case and two conditional preservation cases under fixed learning procedures.
2. **Cross-support characterization:** common-input evidence that the ID2 deficit extends to independent Native synthetic support, together with a support-conditioned ordering of the learned Students.

Synthetic model compression, local operator approximation, and marginal-preserving permutation provide the methodological background. The empirical question here concerns complete position-specific representations, internal residual-update targets, and their learned transfer under controlled recombination.

## 2 Related Work

### 2.1 Synthetic examples and model compression

Learning from a trained model's responses has a long history. TREPAN queries neural networks to extract tree representations [4]. In *Model Compression*, Buciluǎ et al. compare pseudo-data generators for learning an ensemble's function with a smaller neural network. Their RANDOM generator independently samples attributes from empirical marginals; the ensemble relabels the resulting examples. Comparisons with MUNGE show the relevance of joint structure to transfer [3]. This establishes the broader connection between synthetic organization and restricted-model learning.

Our experiment brings that connection to a Transformer hidden-state boundary. Its control preserves a finite multiset of **complete vectors** at every position, rather than drawing input attributes independently. Fresh residual-update labels, matched learning procedures, and common-support evaluation make the consequences of this control directly measurable.

### 2.2 Data-free and layer-wise distillation

Data-free distillation synthesizes supervision from information associated with a trained model. Lopes et al. use model metadata, while DeepInversion optimizes synthetic inputs using Teacher

statistics [9, 18]. Horton et al. generate layer-wise feature inputs for local CNN compression [6]. Ma et al. synthesize continuous pseudo-embeddings for text-classification distillation [10]. These methods establish continuous synthetic inputs and local approximation as natural settings for compression.

Native generation uses a frozen Transformer prefix to construct synthetic sequences from random token IDs. The present experiment keeps the regression objective fixed and changes how its position vectors are grouped. Dependency-aware synthesis is also central to Krukowski et al.'s model-inversion approach to continual learning [8]. Our finite intervention complements such generator studies by controlling exactly which empirical input information is retained.

### 2.3 Permutation and relational distillation

FactorVAE independently permutes latent coordinates across a minibatch to approximate a product of marginals [7]. Q_pos applies the same permutation principle to whole position-specific hidden vectors. The grouping unit determines the invariant: all within-vector information and each position's complete empirical vector distribution remain intact.

FitNets supervises intermediate representations, Relational Knowledge Distillation transfers relations among examples, and MiniLMv2 distills internal attention relations [11, 13, 16]. These approaches alter the supervision being transferred. Here, the full-update loss is constant across arms, and the relation being changed is the grouping of positions within an input example.

### 2.4 Fidelity and generalization

Stanton et al. distinguish Teacher fidelity from generalization and show that optimization and distillation data affect the learned match [14]. Wu et al. examine fidelity under changes in underlying mechanisms [17]. Distribution-shift and underspecification studies likewise motivate evaluating approximations beyond their training distribution [1, 5, 15]. Our value, constructed history-response, and cross-support measurements connect these concerns to a controlled internal-block learning task.

## 3 Problem Setup and Support Intervention

### 3.1 Internal operator learning

Let $G_l(x) \in \mathbb{R}^{L\times d}$ denote embeddings and all frozen blocks preceding zero-based block $l$. The selected Teacher block maps $H$ to $H + T_l(H)$, where $T_l$ is its residual update. A compressed Student $S_\theta$ has the same external width and minimizes

$$\widehat{L}_Q(\theta) = \frac{1}{nLd}\sum_{j=1}^{n} \|S_\theta(H_Q^{(j)}) - T_l(H_Q^{(j)})\|_F^2.$$

The learner includes architecture, initialization, optimizer, minibatch schedule, and terminal budget. Support adequacy is evaluated relative to this fixed procedure.

**Native synthetic support $Q_N$.** Token IDs are sampled iid uniformly with replacement and passed through $G_l$ in evaluation mode. GPT-2 uses all 50,257 embedding-row IDs; Pythia uses its frozen tokenizer-accepted domain. Causal prefix computation couples the resulting vectors. Positional information follows each family's original machinery; GPT-2 position embeddings are not added again at the block boundary.

**Target support $P_l$.** Natural-language windows passed through the same prefix induce the

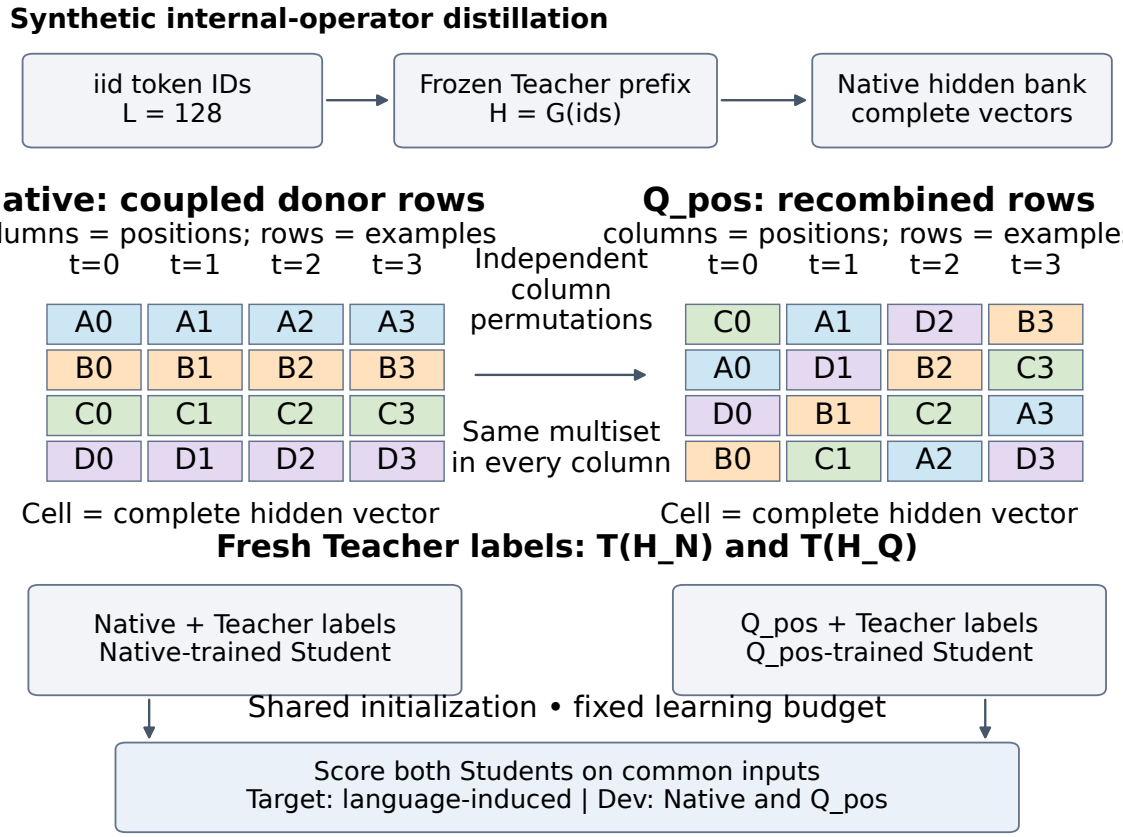



**Figure 1:** The controlled support intervention. Random token IDs pass through a frozen prefix to form Native hidden sequences. Independent row permutations at each position reassign complete vectors while preserving every column multiset. Both banks receive fresh Teacher residual-update labels and train paired Students. Target evaluation uses real-language-induced inputs; exploratory cross-support evaluation uses independent Native and Q_pos dev banks. The four-row example is schematic.

target hidden states. These states evaluate synthetic-to-target transfer. The Teacher is pre-trained on language, while Student training uses the synthetic banks.

### 3.2 Exact finite position-vector preservation

For a Native bank $H_N$ with $n$ rows, independently draw a uniform permutation $\pi_t$ of row indices at each position $t$, and define

$$H_Q[j,t] = H_N[\pi_t(j),t], \qquad Y_Q[j] = T_l(H_Q[j]).$$

For every position,

$$\{H_Q[j,t] : j = 1,\dots,n\} = \{H_N[j,t] : j = 1,\dots,n\}$$

as multisets. This preserves multiplicity, full-vector support, and every statistic determined by the empirical column. The permutations are independent across positions and allow donor coincidences. Train, dev, and audit banks use separate frozen random roles.

Q_pos changes finite row coupling while retaining contextual information inside each vector. It is a finite recombination, rather than an exact population product distribution. Fresh labels incorporate the Teacher's response to the new arrangement. The treatment therefore includes any resulting changes in supervision, approximation, and optimization.

### 3.3 Serial and parallel block updates

Including each branch's normalization and projections, GPT-2 has the serial update

$$T(H) = A_T(H) + M_T(H + A_T(H)),$$

and the selected Pythia block has parallel branches

$$T(H) = A_T(H) + M_T(H).$$

Students follow their respective Teacher topology. A history change at a fixed query can affect GPT-2's MLP through the attention intermediate state. Pythia's MLP reads only the current input vector, so this branch is invariant under that intervention. These structures guide the supplementary response analysis.

## 4 Experimental Protocol

### 4.1 Configurations and training

Table 1 summarizes the formal campaigns using GPT-2 and Pythia Teachers [2, 12]. Within a pair, the two Students share initialization, donor-bank identity, and minibatch indices. All run exactly 10,000 optimizer updates, with terminal checkpoints used for formal evaluation.

**Table 1:** Formal configurations. Block indices are zero-based. Tolerances list Value / Context. Target windows are distinct across campaigns.

| Property | ID2 | XR1 | CMR1 |
|---|---|---|---|
| Teacher / block | GPT-2 Small / 5 | GPT-2 Small / 9 | Pythia-160M / 5 |
| Prefix blocks | 0-4 | 0-8 | 0-4 |
| Topology / activation | serial / gelu_new | serial / gelu_new | parallel / gelu |
| Positions | learned absolute | learned absolute | RoPE |
| Student / Teacher block parameters | 3,546,240 / 7,087,872 | same | same |
| Paired replicates | 48 | 24 | 48 |
| Length / residual width | 128 / 768 | same | same |
| Train / dev / audit rows | 4096 / 256 / 512 | same | same |
| Target windows per environment | 1024 | 1024 | 1024 |
| Registered tolerance | absolute 0.020 / 0.015 | 5% of Native recovery | absolute 0.020 / 0.015 |

Teachers have 12 attention heads of width 64 and an MLP width of 3072. Students have six heads of width 64, internal attention width 384, MLP width 1536, and residual width 768. Projection weights are independently initialized with standard deviation 0.02, biases are zero, normalization scales/biases are one/zero, and dropout is disabled.

The shared recipe uses AdamW (betas 0.9/0.999, epsilon $10^{-8}$, weight decay 0), peak learning rate $10^{-3}$, 200-step warmup, cosine decay to $10^{-4}$, gradient clipping at 1, and effective batch 32 from four microbatches of eight. The deterministic FP32 configuration disables AMP and TF32 and preserves family-specific masks and positional interfaces.

### 4.2 Value and history-response endpoints

For environment $k$, Value is

$$R_k(S) = \frac{\sum_{H\in P_k} \|S(H) - T(H)\|_F^2}{\sum_{H\in P_k} \|T(H)\|_F^2}, \qquad R(S) = \frac{1}{3}\sum_{k=1}^{3} R_k(S).$$

The C4, peS2o, and StackExchange environments receive equal weight after separate normalization. A zero-update predictor has Value 1.

Context measures fidelity to a constructed history replacement. At queries $t \in \{63, 127\}$, earlier positions $[0, t)$ are replaced by a frozen independent nonself donor; the query and later positions are fixed. Define $\Delta T_t = T(H')_t - T(H)_t$ and similarly $\Delta S_t$. The formal endpoint is

$$C(S) = \frac{1}{3}\sum_{k} \frac{1}{2} \sum_{t\in\{63,127\}} \frac{\sum_{H\in P_k} \|\Delta S_t - \Delta T_t\|_2^2}{\sum_{H\in P_k} \|\Delta T_t\|_2^2}.$$

Context thus evaluates a controlled functional response, including potentially off-manifold splices, rather than ordinary semantic understanding. A deterministic token-wise Student has Context 1 when the response denominator is nonzero. Pythia's fixed-query MLP response cancels; GPT-2's serial MLP can participate. Earlier Silent Teacher/ID1 Context scores pooled the two queries before division and are reported separately in Appendix B.

For each endpoint, let $N_i, Q_i$ be the Native- and Q_pos-trained Student scores, and $D_i = Q_i - N_i$. Positive D favors Native. Paired replicates are the inferential units. Bounds condition on the realized frozen target ledger and exclude corpus-sampling uncertainty.

### 4.3 Registered decision rules

The one-sided bounds are $U_{95}(X) = \bar{X} + t_{.95,m-1}s_X/\sqrt{m}$ and $L_{97.5}(X) = \bar{X} - t_{.975,m-1}s_X/\sqrt{m}$. Each campaign retains its own rule:

- **ID2:** absolute margins $\delta_R = 0.020$, $\delta_C = 0.015$. Either $L_{97.5}(D) > \delta$ establishes material loss; both $U_{95}(D) < \delta$ establish preservation. Native upper 95% bounds are checked against historical reference thresholds 0.674317 / 0.721632, with the registered material-loss precedence.
- **XR1:** $G_i = 1 - N_i$ and $Z_i = D_i - 0.05G_i$. Native requires both $L_{95}(G) > 0.20$. Conditional on qualification, both $U_{95}(Z) < 0$ establish relative preservation; either $L_{97.5}(Z) > 0$ establishes relative loss. Pair-level Z retains the variance of recovery and penalty together.
- **CMR1:** Native also requires both $L_{95}(G) > 0.20$, while loss/preservation use ID2's absolute margins. Preservation under qualification yields its qualified terminal outcome.

The 97.5% lower bounds protect an either-endpoint loss claim across two endpoints; preservation requires both upper-bound conditions. Across campaigns, comparisons are descriptive because blocks, models, supports, ledgers, and in some cases decision scales differ. There is no registered common cross-campaign significance test.

### 4.4 Research sequence and evaluation identity

Early Silent Teacher work used WikiText and PG-19 development/validation data, followed by a held-out PG-19 confirmation. These outcomes informed the later intervention sequence. ID1, ID2, XR1, and CMR1 then froze separate construction rules, random roles, target ledgers, and scientific decisions. Each formal Q_pos ledger contains 3072 length-128 windows, with predecessor exposure excluded from the three known source families. Registration here refers to the archived campaign-specific scientific freeze.

Execution amendments and exposure boundaries are documented in Appendix A, including a feasibility evaluator repair after Value outcomes, ID1 repeat evaluation, ID2 analysis recovery, XR1 orchestration repair, and CMR1 prefreeze control and metadata-access incidents. The appendix identifies the relevant scientific conditions and retained result identities.

### 4.5 Exploratory common-support evaluation

After the formal results, both terminal Students were scored on both original independent dev banks (256 rows, length 128). Each support column supplies identical hidden inputs, direct Teacher labels, and a common Teacher-energy denominator to both Students. All 480 cells across 120 pairs were recomputed, with historical own-support scores used as identity checks. Same-column paired comparisons are central because the two supports have different denominators. This post-outcome diagnostic complements the formal target results.

## 5 Results

### 5.1 Feasibility and the earlier intervention

Silent Teacher established partial synthetic-to-target fidelity at GPT-2 block 5 in three held-out PG-19 seeds. Native target Value was 0.666047-0.698391; the tested Gaussian boundary control scored 2.895546-3.213844 despite own-audit Value 0.267775-0.287701. Native also had lower Context and predictive KL than Gaussian. These results motivate examining the organization of Teacher-derived synthetic inputs. Appendix B retains all seeds and the contrasting token-wise and parameter-copy controls.

ID1 compared coherent upstream token-half splicing with composite separately computed hidden halves. Across 24 pairs, its mean Value penalty was 0.003801 (SD 0.005386), below the registered 0.05 materiality threshold. Q_pos subsequently introduced exact finite column control, whereas ID1 preserved a construction-level sampling law. Their different controls motivate treating the two interventions separately.

### 5.2 Formal Q_pos outcomes

Table 2 reports all formal continuous effects, and Table 3 gives the decision evidence. Preservation cases have positive average penalties within their registered tolerances.

**Table 2:** Formal target fidelity. N/Q are mean Student scores; D is the paired Q-minus-N penalty. SD(D) is across replicate pairs. All values are normalized under the endpoint definitions in Section 4.2.

| Study | Endpoint | Pairs | N | Q | Mean D | SD(D) | D > 0 |
|---|---|---|---|---|---|---|---|
| ID2 | Value | 48 | 0.659816 | 0.723623 | 0.063807 | 0.015806 | 48/48 |
| ID2 | Context | 48 | 0.710719 | 0.740457 | 0.029738 | 0.024408 | 43/48 |
| XR1 | Value | 24 | 0.487372 | 0.500135 | 0.012763 | 0.007231 | 24/24 |
| XR1 | Context | 24 | 0.605316 | 0.608207 | 0.002891 | 0.028016 | 15/24 |
| CMR1 | Value | 48 | 0.314554 | 0.322472 | 0.007917 | 0.008034 | 40/48 |
| CMR1 | Context | 48 | 0.412340 | 0.414828 | 0.002488 | 0.043333 | 28/48 |

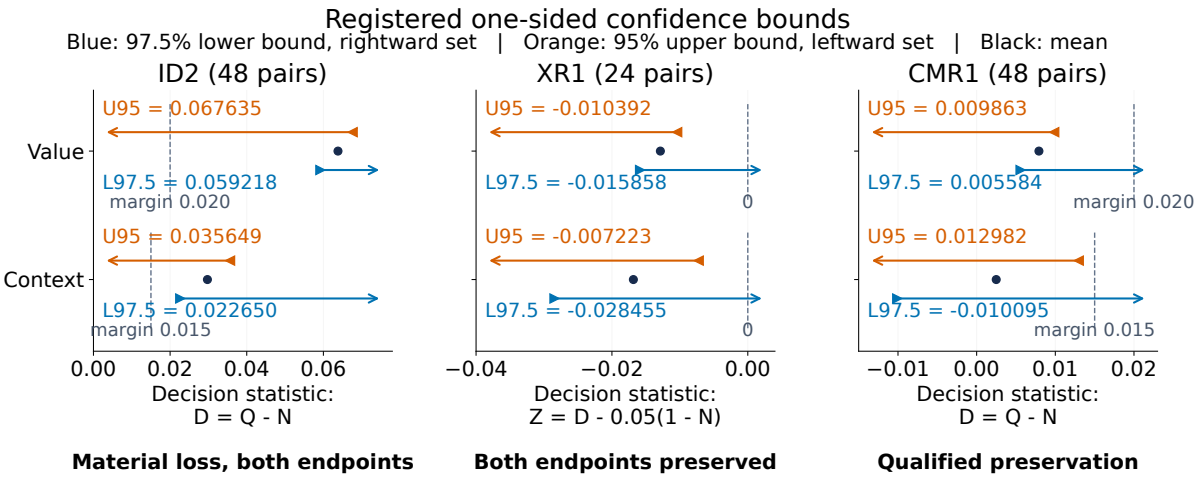



**Figure 2:** Formal decisions on their registered scales. Each column is one campaign. Black points show the mean decision statistic. Blue rightward arrows start at its one-sided 97.5% lower bound; orange leftward arrows start at its one-sided 95% upper bound. The two arrows represent separate one-sided confidence sets, not a combined two-sided interval. Dashed lines mark the decision boundary: D margins 0.020 / 0.015 for ID2/CMR1 (Value / Context), and zero for XR1's Z = D - 0.05(1 - N). Material loss requires a lower bound above the boundary for either endpoint; preservation requires upper bounds below it for both endpoints, with Native qualification satisfied as shown in Table 3. All means and bounds are copied from the existing formal records. The column-specific scales preserve the different registered questions.

**Table 3:** Registered decision evidence. Entries give Value / Context. Loss uses one-sided 97.5% lower bounds; preservation uses one-sided 95% upper bounds. Qualification is a separate requirement. XR1's decision variable is Z; ID2/CMR1 use D.

| Study | Native qualification | Loss / preservation bounds | Terminal outcome |
|---|---|---|---|
| ID2 | U95(N) = 0.662511 / 0.716535 < 0.674317 / 0.721632 | L97.5(D) = 0.059218 / 0.022650 > margins 0.020 / 0.015; U95(D) = 0.067635 / 0.035649 | Material loss, both endpoints |
| XR1 | L95(G) = 0.510848 / 0.388770 > 0.20 | U95(Z) = -0.010392 / -0.007223 < 0; L97.5(Z) = -0.015858 / -0.028455 | Both endpoints preserved |
| CMR1 | L95(G) = 0.684391 / 0.580793 > 0.20 | U95(D) = 0.009863 / 0.012982 < margins 0.020 / 0.015; L97.5(D) = 0.005584 / -0.010095 | Qualified preservation |

ID2's loss exceeds both absolute margins. Complete position-wise vector collections therefore do not preserve target fidelity adequately for this block-5 learner under the fixed budget. XR1 passes qualification and relative preservation; its mean penalty is 2.49% of mean Native recovery for Value and 0.73% for Context. CMR1 qualifies and passes absolute preservation in another Teacher family. The combination of outcomes makes the empirical dependence on row organization configuration-specific.

## 5.3 Cross-support characterization

Common-support scoring reveals how the two Students' approximation quality depends on their inputs (Table 4). The diagonal scores reproduce historical dev values within $10^{-9}$, while the off-diagonal scores expose behavior that own-support evaluation cannot show.

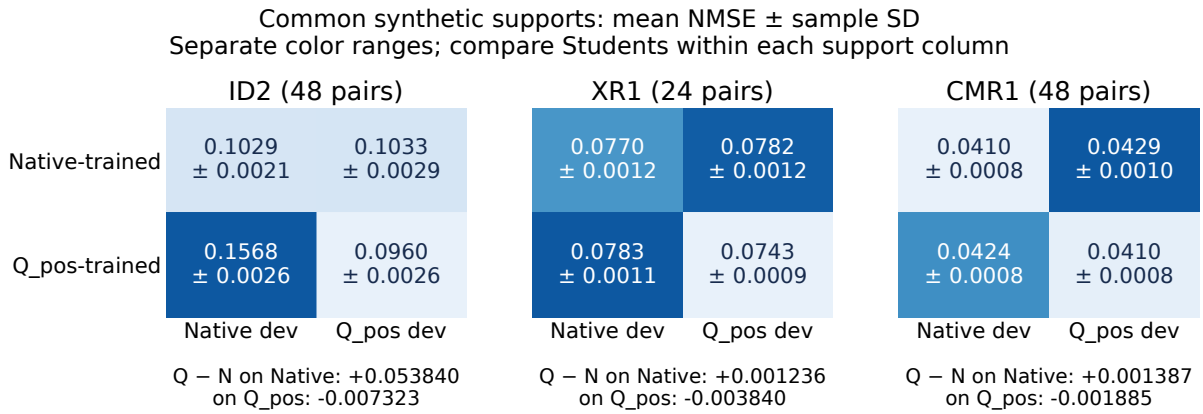



**Figure 3:** Common-support Student ordering. Rows identify the Student's training support; columns identify its independent evaluation support. Cells give mean NMSE ± sample SD, and the lower annotations give paired Q-trained-minus-N-trained differences by column. Panel colors use separate ranges. Within-column comparisons share a Teacher-energy denominator. The matrix is an exploratory synthetic diagnostic using all original formal replicates.

**Table 4:** Exploratory common synthetic-support matrix. Mean ± sample SD across all formal pairs. Students in a column share inputs, Teacher labels, and denominator; Native and Q_pos columns have different Teacher energies.

| Study | Student | Native dev | Q_pos dev |
|---|---|---|---|
| ID2 | Native-trained | 0.102917 ± 0.002054 | 0.103293 ± 0.002937 |
| ID2 | Q_pos-trained | 0.156757 ± 0.002625 | 0.095970 ± 0.002594 |
| XR1 | Native-trained | 0.077021 ± 0.001166 | 0.078161 ± 0.001187 |
| XR1 | Q_pos-trained | 0.078257 ± 0.001059 | 0.074322 ± 0.000915 |
| CMR1 | Native-trained | 0.041004 ± 0.000847 | 0.042914 ± 0.001015 |
| CMR1 | Q_pos-trained | 0.042392 ± 0.000814 | 0.041029 ± 0.000795 |

On Native dev support, ID2's paired Q-minus-N difference is +0.053840 (SD 0.002857), positive in all 48 pairs. On Q_pos dev support it is -0.007323 (SD 0.003491), positive in only one pair. The Q_pos Student's deficit consequently extends to independent Teacher-derived synthetic inputs. It cannot be described solely as a failure that emerges when moving to language-induced states.

XR1 and CMR1 also reverse their mean Student ordering across support columns. Their Native-column penalties are +0.001236 and +0.001387, and their Q_pos-column differences are -0.003840 and -0.001885. ID2's Native-column discrepancy is descriptively larger. These common-input comparisons establish support-conditioned approximations; differences in Teacher and scale remain part of the cross-configuration comparison.

Own-support diagonals compare different input-label distributions. The matrix instead answers which Student better approximates the Teacher on the same inputs. The ID2 Native-dev penalty +0.053840 and its formal target Value penalty +0.063807 are measurements on different distributions, rather than terms in a numerical loss decomposition.

### 5.4 Functional characterization

Supplementary donor-aligned measurements hold the query vector and position fixed while replacing its historical combination. ID2's Q_pos Student has relative Teacher-response error 0.477392 versus 0.252638 for the Native Student; XR1 and CMR1 lack this Q_pos disadvantage. ID2 has the highest relative Teacher response energy, while XR1 has the highest raw

energy. Error decompositions reveal substantial branch compensation, and fixed short continuation modifies response reproduction and final error in partially different ways. Appendices D-E give these exploratory measurements and their algebraic definitions.

# 6 Discussion

## 6.1 Context inside vectors and organization across vectors

Q_pos retains all empirical information available within a single position vector. ID2 nevertheless incurs material target penalties, showing that the downstream learning procedure benefits from Native cross-position organization beyond that retained information. XR1 and CMR1 show that the size of this benefit is not uniform: their approximations meet the corresponding preservation criteria. Together, the results make the practical question sharper than the abstract observation that marginals leave a joint distribution unspecified. What matters empirically is how much the changed joint organization affects a given learner's fidelity.

The treatment changes input arrangement and recomputed labels together. Its measured effect belongs to the complete support-and-supervision construction under the fixed learner. Moving between campaigns also changes the block, prefix-induced support, and, for CMR1, model family and Student topology. The preservation cases therefore describe tested configurations rather than isolate a single architectural cause.

## 6.2 Common support locates the discrepancy

Cross-support scoring changes the interpretation of ID2. The Q_pos Student is relatively successful on its own independent support, but substantially worse on Native synthetic inputs. This places part of the approximation problem before the additional shift to natural-language-induced states. It also explains why diagonal dev scores offer an incomplete picture: two learned functions can have favorable scores on different supports and reverse order under common-input evaluation.

This characterization focuses future explanation on the relationship between training supervision and the behaviors required on Native support. Finite optimization, inductive bias, capacity, and support-specific constraints can all affect that relationship. The current evidence identifies where the approximation differs more clearly than why it differs.

## 6.3 What the supplementary diagnostics resolve

Teacher history sensitivity, Student response reproduction, and final output error measure different aspects of the learning problem. The response defect gives a functional description of ID2's Q_pos approximation. The exact decompositions show why attributing total error to one positive branch norm is misleading: negative inner products offset substantial branch errors, and ID2's net propagation-associated change is negative despite an increased propagation norm.

Short continuation adds a local intervention on learning. Extra pure Q_pos training modestly improves the historical Q Student on Native audit; a 10% Native loss weight improves it more relative to the same-start pure continuation. Starting-point differences persist after 1000 matched-compute updates. Moreover, the Native-start mixed branch improves response reproduction while slightly worsening Native audit from baseline. Functional improvement and final-error improvement are thus separable, and these diagnostics do not identify a causal mediator. Appendix F separates Teacher history sensitivity from support dependence in a restricted-Student example.

## 7 Limitations

The study covers three formal configurations, two Teacher families, length 128, and one compressed width recipe under finite budgets. It establishes configuration-specific outcomes, with convergence and broader architecture coverage remaining open. Normalized fidelity measures depend on their Teacher energies and aggregation; raw and relative response energies answer different scaling questions.

Target inference conditions on fixed ledgers from known source families. Early development informed subsequent campaign choices, and fresh window identities do not establish domain novelty or absence from Teacher pretraining. StackExchange may overlap Pythia's pretraining sources. The independent synthetic matrices, functional diagnostics, and continuation were conducted after formal outcomes and support descriptive interpretation. Their recorded supports, positions, and scheduled time points are retained in the supplements.

## 8 Conclusion

Complete finite position-vector marginal preservation can coexist with substantial losses in Transformer-block learning and transfer. ID2 demonstrates that failure under a fixed compressed learner, while XR1 and CMR1 provide conditional preservation cases. Cross-support evaluation further exposes a Native synthetic deficit and a reversal of Student ordering across supports. The combination makes synthetic support organization an empirically consequential, configuration-dependent aspect of internal-operator distillation and establishes common-support evaluation as a clear way to characterize the learned approximations.

## Generative AI assistance

Generative AI tools, including OpenAI Codex and other language-model assistants, substantially assisted research planning, experimental and analysis code development, literature organization, and manuscript drafting and revision. The author retains responsibility for the scientific content, verification of results and references, and the final manuscript.

## Code and data availability

A curated software and results archive, with the manuscript sources, is prepared for release under the MIT license (original software) and CC BY 4.0 (manuscript and result summaries). At this version's preparation date it is not publicly released; no public repository link or archival DOI has been assigned. The archive supports figure regeneration and inspection of recorded results; model weights, language corpora, target arrays, and historical Student checkpoints are not redistributed. Appendix A identifies the existing result versions and scientific provenance.

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# A Research sequence and provenance

## A.1 Development and scientific freezes

Silent Teacher progressed from a token-wise MLP task to full-block learning, synthetic qualification, WikiText/PG-19 development, and held-out PG-19 confirmation. Its confirmation set is historically spent. Subsequent ID1, ID2, XR1, and CMR1 studies each froze construction rules, random roles, target selection, and scientific decision criteria after the preceding evidence became available.

Each formal Q_pos target ledger contains 1024 length-128 windows from each of C4, peS2o, and StackExchange. Deterministic predecessor-exposure exclusions preserve the campaign-specific retained identities. The ledgers differ; Pythia also uses a different tokenizer. A frozen record's NOT_RUN state refers to the freeze date, with completed results recorded separately.

Pythia-160M uses revision `50f5173d932e8e61f858120bcb800b97af589f46`, weight SHA-256 `29d2e457a664e41c12c735f20a36dc0956a665f614a54ce5db21a32e75965270`. Its parallel residual configuration uses RoPE fraction 0.25/base 10000 and LayerNorm epsilon 1e-5. All formal Students are terminal checkpoints at 10,000 updates. Original receipts identify Teacher weights, Student parameters, construction roles, and input hashes.

## A.2 Existing evidence versions

The following Git versions identify the records used here. Result versions and content hashes identify the evidence; the curated public archive is prepared but remains unpublished at this version's preparation date.

| Evidence | Git version |
|---|---|
| Silent Teacher confirmation results | `0d6a0fe7f5ebff429c0861264a9c2046d5856c04` |
| ID1 report / erratum | `f6ee4f9409f32824134fcaf7bda0a6ad9183b4c3` / `e23ffc010911083ddd119747ea35a80d11856735` |
| ID2 formal results | `fd8b2f073b5dc24244c2615126ed52fffe65c996` |
| XR1 formal results | `a603c6e1b62532b53902209da0a29ed12582bec8` |
| CMR1 comparison / evaluation implementation | `6e1cddecfd47fd0f613ff6eb50c7974ffbdfe925` / `1485bbe52767c02fd93e48d10564779b138422e1` |
| Training-support diagnostics | `56fe0fc9fe1c10d45f7688dd9517b099a593dc2a` |
| Cross-support diagnostics | `91e280115b8b21b7277daa8678efabd326ab30dd` |
| Functional-response diagnostics | `eb0338f76ca823df07c2e7c65d07b67e87c6c015` |
| Short continuation | `13bec5abe3d6a14286e0966cf7218339e6ab486c` |

Formal aggregate-result SHA-256 identities are:

| Campaign | SHA-256 |
|---|---|
| ID2 | `1ce0085a2eb82b16e15ed1cbbee21fd6ea2944ac6911c0ba120b1c3287decd7f` |
| XR1 | `529b1768933d910974319e2fa7719362e9ee933146d2af8f031c695156d612c5` |
| CMR1 | `0fcd19b492d95df1a5010019df3b80319e520940f7c12c6ba305b6ec541552cc` |

## A.3 Execution and exposure boundaries

**Silent Teacher and early qualification.** The early B0 numerical qualification failed ($0.05000536 > 0.05$), followed by a procedural waiver and deterministic selection repair.

Later full-block confirmation is separate evidence. After held-out Value outcomes, the Context/causality evaluator required a memory-bounded streaming repair. Equivalence was checked before use on 1600 realized windows (maximum difference 1.57e-13, tolerance 1e-12); data, donors, checkpoints, definitions, and criteria were unchanged.

**ID1.** A secondary-key failure required a repair and repeat evaluation. Both executions and a later wording erratum remain in the historical record. The NLL point-estimate direction agreed with Value/KL; Context differed.

**ID2.** A gate bug launched both replicate-01 arms before the intended execution tag. Outcomes were not inspected, and unchanged-seed replay was bit-identical. A pre-outcome driver completed the evaluator stub. Following target evaluation, an analysis-persistence failure was recovered from the same immutable metrics, with 41 bit-exact comparisons and zero mismatches.

**XR1.** Training-control incidents required gate repairs. Evaluation v8 aborted after transient first-Student/first-environment metrics that were neither persisted nor exposed as scientific outcomes. Repaired v9 retained all 48 valid Student records without recomputing valid persisted records.

**CMR1.** Qualification and RNG-control revisions occurred before the scientific freeze. An overbroad review search exposed historical target metadata but no example bodies, labels, model outputs, or target scores. The archived assessment records a procedural access violation without assessed outcome contamination; canonical selection remained unchanged. An alternative unauthorized qualification root was transient and not adopted. The production Teacher wrapper was repaired before formal training.

These events qualify the execution history and observed exposure, while superseded runs remain outside the formal replicate counts.

# B Feasibility and earlier intervention

## B.1 Historical feasibility and endpoint definitions

The held-out PG-19 confirmation used 3196 windows and 405,892 scored tokens with seeds 501-503. Native passed the original Value, pooled-query Context, and predictive-KL criteria in all three seeds, yielding HELDOUT_CONFIRMED_REPLICATED. Context pooled the two query numerators and denominators within the environment before division, unlike the equal-query formal endpoint used in ID2/XR1/CMR1. Lower KL indicates closer whole-model predictions after block replacement; NLL measures language-model quality and can move differently.

**Table B1:** Complete historical feasibility scores. Lower is better. The Gaussian own-audit column evaluates its independent synthetic support; the other columns evaluate the held-out target. Values use the persisted precision, rounded to six decimals.

| Seed | Native Value | Native Context | Native KL | Gaussian own-audit Value | Gaussian target Value | Gaussian target Context | Gaussian KL |
|---|---|---|---|---|---|---|---|
| 501 | 0.698391 | 0.756193 | 0.115437 | 0.267775 | 2.895546 | 1.117745 | 1.033124 |
| 502 | 0.672299 | 0.702185 | 0.119684 | 0.287235 | 3.073787 | 1.166020 | 1.046195 |
| 503 | 0.666047 | 0.729700 | 0.115233 | 0.287701 | 3.213844 | 1.166658 | 1.080099 |

TokenWise controls have Context 1 and target Value 0.724212-0.730071. The parameter-copy control has Value 0.868789 and Context 0.690712, illustrating different fidelity profiles: it has a lower Context error than Native but worse Value and predictive KL. These controls establish partial transfer rather than exact operator recovery.

### B.2 Earlier intervention and terminal fitting

For ID1, let $z^A, z^B$ be iid token-ID donors and $c = 64$. The coherent and composite arms are

$$H^+ = G(z^A_{<c}, z^B_{\geq c}), \qquad H^- = (G(z^A)_{<c}, G(z^B)_{\geq c}).$$

Causality makes their first-half hidden states identical. Their second halves arise from different preceding token contexts. Both receive fresh full-block Teacher labels. This is not a permutation of one finite hidden bank, so ID1's small effect cannot be treated as a lesser dose of the precisely controlled ID2 intervention.

The later exploratory training-support review covered 240 formal arm curves across ID2/XR1/CMR1. Terminal own-support dev means (Native / Q_pos) were 0.102917 / 0.095970, 0.077021 / 0.074322, and 0.041004 / 0.041029. These compare different supports and Teacher-energy denominators. Curves remained declining near their terminal region; finite stable records do not establish convergence. Neither lower diagonal NMSE nor continued decline alone identifies the source of ID2's cross-support deficit.

## C Same-column cross-support differences

**Table C1:** Paired Q_pos-trained-minus-Native-trained error on each common dev support. Ranges are replicate minima and maxima, not intervals for an expectation. All original formal pairs are included.

| Campaign | Support column | Mean difference | Sample SD | Range | Positive pairs |
|---|---|---|---|---|---|
| ID2 | Native | +0.053840 | 0.002857 | +0.048937 to +0.061634 | 48/48 |
| ID2 | Q_pos | −0.007323 | 0.003491 | −0.015187 to +0.001565 | 1/48 |
| XR1 | Native | +0.001236 | 0.001662 | −0.002881 to +0.004067 | 19/24 |
| XR1 | Q_pos | −0.003840 | 0.001558 | −0.007695 to −0.001151 | 0/24 |
| CMR1 | Native | +0.001387 | 0.001033 | −0.000756 to +0.004257 | 42/48 |
| CMR1 | Q_pos | −0.001885 | 0.001221 | −0.004251 to +0.001672 | 2/48 |

Per-replicate records preserve raw SSE, Teacher energy, bank and checkpoint identities. Both Students share the same denominator within a column. GPT-2 Teacher scoring uses the historical batch size 32 and Student scoring size 8; CMR1 uses size 8 for both. The entire recomputed diagonal agreed with historical terminal dev values within 1e-9.

## D Donor-aligned response and exact error geometry

### D.1 Definitions and negative controls

At Q_pos row $j$, position $t$, the Native donor is $\pi_t(j)$. Define

$$\delta T_{jt} = T(H_Q[j])_t - T(H_N[\pi_t(j)])_t, \quad \delta S_{jt} = S(H_Q[j])_t - S(H_N[\pi_t(j)])_t.$$

The query vector is bitwise identical, positions and masks match, and changed future positions cannot affect a causal query. This differs from the formal target Context splice: all positions of

the synthetic row are recombined, with the query compared to its own position-specific donor. Both are functional responses, but they use different interventions and supports.

Because the paired query input is identical, its residual-update response equals its full-block-output response. On identical inputs, Student–Teacher update error likewise equals block-output error. Nevertheless, response reproduction and final error remain distinct measurements: for one Student, $\delta S - \delta T = e(H_Q[j])_t - e(H_N[\pi_t(j)])_t$. Thus response reproduction is a paired difference of its errors on two supports, not independent evidence of a causal mechanism.

For each replicate the public denominator is $E_N = \sum_{jt} \|T(H_N[j])_t\|^2$. Both Students, both supports, and all decomposition terms share it. Position-specific summaries use the corresponding Native energy $E_{N,t}$; group summaries pool numerators and denominator before dividing. Reported values average replicate-level ratios. Raw response energy is the squared norm per query vector, not per scalar coordinate. Relative response reproduction error is $\sum \|\delta S - \delta T\|^2 / \sum \|\delta T\|^2$. Gain is $\langle \delta S, \delta T \rangle / \|\delta T\|^2$ and cosine uses all response vectors flattened together, rather than averaging local cosines.

At $t = 0$, the same Teacher or Student has zero response difference because there is no earlier history. For Pythia, the directly query-dependent MLP branch has zero response difference at every position. Both references were exactly zero in the completed batch. They do not require equality of outputs from two *different* Students. Checks covered input identity, permutation bijections, labels, interfaces, masks, positional machinery, and algebraic closure.

The initial GPT-2 smoke used Teacher batch size 8 and failed the label-identity check (maximum difference 1.0681e-4). Restoring historical size 32 yielded exact labels without relaxing tolerance. FP32 forwards and stored update sums were retained; differences and inner products used FP64. Vector closure tolerance was 2e-5, relative SSE closure 1e-6, and the two-Student identity tolerance 1e-12. Small FP32 accumulation residues were recorded, not discarded. The archived suite reports 121 passing tests after this repair. Complete position summaries include all 128 positions plus the fixed t=0/63/127 references; no favorable query positions were selected after inspection.

## D.2 Teacher intervention strength and Student responses

**Table D1:** Teacher response energy (mean ± sample SD). Raw energy is per 768-dimensional query vector. The normalized value divides pooled energy by the corresponding Native Teacher-update energy. This is neither formal target Context nor the Cross-Support Q_pos-column NMSE denominator.

| Campaign | Position group | Raw energy | Relative to Native Teacher energy |
|---|---|---|---|
| ID2 | all positions | 80.771 ± 0.498 | 0.201133 ± 0.001169 |
| ID2 | 63 | 79.983 ± 2.401 | 0.244596 ± 0.007234 |
| ID2 | 127 | 64.436 ± 2.365 | 0.199660 ± 0.007138 |
| XR1 | all positions | 268.047 ± 2.572 | 0.132458 ± 0.001227 |
| XR1 | 63 | 274.961 ± 11.068 | 0.131109 ± 0.004582 |
| XR1 | 127 | 218.677 ± 10.492 | 0.103969 ± 0.005159 |
| CMR1 | all positions | 15.868 ± 1.042 | 0.116186 ± 0.007369 |
| CMR1 | 63 | 16.725 ± 1.322 | 0.116462 ± 0.008057 |
| CMR1 | 127 | 16.596 ± 1.136 | 0.109945 ± 0.007006 |

ID2 has the largest normalized response strength, while XR1 has the largest raw strength. These answer different scaling questions and do not determine Student learning difficulty.

**Table D2:** Overall donor-aligned response reproduction. Error is relative to Teacher response energy; gain and cosine have the definitions above. Parentheses give sample SDs. Numbers are not target Context scores.

| Campaign | Student | Relative response error | Projection gain | Cosine |
|---|---|---|---|---|
| ID2 | Native | 0.252638 (0.019425) | 0.748623 (0.016884) | 0.864443 (0.011248) |
| ID2 | Q_pos | 0.477392 (0.020856) | 0.521098 (0.019338) | 0.722778 (0.014662) |
| XR1 | Native | 0.185215 (0.011385) | 0.813856 (0.011602) | 0.902636 (0.006353) |
| XR1 | Q_pos | 0.176972 (0.011691) | 0.821584 (0.011112) | 0.907190 (0.006442) |
| CMR1 | Native | 0.187793 (0.012579) | 0.808871 (0.011725) | 0.901214 (0.007007) |
| CMR1 | Q_pos | 0.184643 (0.015409) | 0.816821 (0.014254) | 0.902940 (0.008571) |

At the preselected positions 63 / 127, ID2 relative errors are Native 0.246924 / 0.272729 and Q_pos 0.466508 / 0.537749. XR1 values are Native 0.170322 / 0.176229, Q_pos 0.163245 / 0.165972; CMR1 values are Native 0.188454 / 0.202034, Q_pos 0.186832 / 0.199748. The Q_pos disadvantage in ID2 is not confined to the pooled result. At t=0, response errors are zero and relative error/cosine are undefined, not assigned zero as if they were measured ratios.

### D.3 Serial and parallel decompositions

For GPT-2 define

$$\begin{aligned} a &= A_S(H) - A_T(H), \\ m &= M_S(H + A_T(H)) - M_T(H + A_T(H)), \\ p &= M_S(H + A_S(H)) - M_S(H + A_T(H)). \end{aligned}$$

Then $e = S(H) - T(H) = a + m + p$, and

$$\|e\|^2 = \|a\|^2 + \|m\|^2 + \|p\|^2 + 2\langle a, m\rangle + 2\langle a, p\rangle + 2\langle m, p\rangle.$$

Here $m$ evaluates the Student MLP at the Teacher attention intermediate state for diagnosis; it does not replace the Student's actual forward pass. $p$ measures how the Student's attention deviation changes its subsequent MLP input and output. This is an exact algebraic decomposition in real arithmetic, not a unique allocation of causal loss.

For Pythia, $a = A_S(H) - A_T(H)$, $m = M_S(H) - M_T(H)$, and $e = a + m$, so the norm expansion has two squared terms and one doubled inner product. There is no serial propagation term.

**Tables D3–D5. Full-output error decomposition.** Each entry is *raw pooled SSE or doubled inner product in millions; normalized value using E_N.* Columns denote Student → input support. Values are replicate means, not ratios of mean raw energies. Cross terms can be negative; squared norms cannot be summed alone to recover full error. Rounding and archived FP32 summation residues explain small displayed closure discrepancies.

**Table D3:** ID2.

| Term | N → N | Q → N | N → Q | Q → Q |
|---|---|---|---|---|
| | 4.014204 | 4.523972 | 3.685635 | 3.430739 |
| attention | 0.305056 | 0.343792 | 0.280087 | 0.260713 |
| squared | | | | |
| | 4.511930 | 4.340246 | 4.365149 | 4.109671 |
| fixed-state MLP | 0.342874 | 0.329831 | 0.331721 | 0.312308 |
| squared | | | | |

| Term | N → N | Q → N | N → Q | Q → Q |
|---|---|---|---|---|
| | 0.593324 | 0.680148 | 0.584322 | 0.561473 |
| propagation squared | 0.045091 | 0.051686 | 0.044406 | 0.042668 |
| | −6.615998 | −6.155106 | −6.224216 | −5.796474 |
| $2\langle a, m\rangle$ | −0.502770 | −0.467748 | −0.472997 | −0.440494 |
| | −0.901920 | −1.139152 | −0.710807 | −0.639745 |
| $2\langle a, p\rangle$ | −0.068548 | −0.086566 | −0.054025 | −0.048616 |
| | −0.247270 | −0.187353 | −0.401265 | −0.458919 |
| $2\langle m, p\rangle$ | −0.018786 | −0.014238 | −0.030489 | −0.034875 |
| | 1.354269 | 2.062756 | 1.298818 | 1.206745 |
| full error | 0.102917 | 0.156757 | 0.098703 | 0.091705 |

**Table D4:** XR1.

| Term | N → N | Q → N | N → Q | Q → Q |
|---|---|---|---|---|
| | 15.746770 | 16.484172 | 15.231714 | 15.596392 |
| attention squared | 0.237479 | 0.248592 | 0.229709 | 0.235204 |
| | 18.035616 | 17.169850 | 17.883137 | 16.957251 |
| fixed-state MLP squared | 0.271982 | 0.258926 | 0.269685 | 0.255721 |
| | 0.970747 | 1.053769 | 0.971983 | 1.000253 |
| propagation squared | 0.014641 | 0.015893 | 0.014659 | 0.015085 |
| | −27.782527 | −27.461882 | −27.140661 | −26.804228 |
| $2\langle a, m\rangle$ | −0.418978 | −0.414138 | −0.409298 | −0.404221 |
| | −2.243989 | −3.537007 | −2.017098 | −3.161020 |
| $2\langle a, p\rangle$ | −0.033855 | −0.053350 | −0.030431 | −0.047679 |
| | 0.380652 | 1.480429 | 0.104187 | 1.197471 |
| $2\langle m, p\rangle$ | 0.005752 | 0.022334 | 0.001581 | 0.018066 |
| | 5.107270 | 5.189330 | 5.033263 | 4.786119 |
| full error | 0.077021 | 0.078257 | 0.075905 | 0.072176 |

**Table D5:** CMR1 parallel block.

| Term | N → N | Q → N | N → Q | Q → Q |
|---|---|---|---|---|
| | 0.953070 | 1.036619 | 0.959827 | 1.031419 |
| attention squared | 0.213000 | 0.231667 | 0.214511 | 0.230505 |
| | 1.026059 | 1.104260 | 1.026059 | 1.104260 |
| MLP squared | 0.229313 | 0.246789 | 0.229313 | 0.246789 |
| | −1.795646 | −1.951186 | −1.792999 | −1.951270 |
| $2\langle a, m\rangle$ | −0.401309 | −0.436064 | −0.400719 | −0.436082 |
| | 0.183483 | 0.189693 | 0.192887 | 0.184409 |
| full error | 0.041004 | 0.042392 | 0.043105 | 0.041211 |

For ID2 on Native support, the normalized Q-minus-N changes are +0.038736 in attention squared, −0.013043 in fixed-state MLP squared, +0.006596 in propagation squared, and +0.035022/−0.018018/+0.004548 in the three cross terms. The *net propagation-associated* change, including both cross terms involving p, is −0.006874, despite its positive squared-term change. This prevents attributing the total extra error to a standalone propagation loss. CMR1's same-Student MLP squared error is identical across supports by the full-vector column invariant, but different Students can have different MLP errors. Their error compensation differs as well.

The Q_pos full-error numbers in these tables are not the Q_pos column of Table 4. Here both supports use E_N; Table 4 uses each column's own Teacher energy. For example, ID2's Q-column difference is −0.006998 here and −0.007323 in Cross-Support. Both are valid under their declared denominators.

### D.4 Output disagreement and direction

Let $e_N = S_N - T$ and $d = S_Q - S_N$ on the same input. Exactly,

$$\|S_Q - T\|^2 - \|S_N - T\|^2 = \|d\|^2 + 2\langle e_N, d\rangle.$$

**Table D6:** Two-Student identity on pooled dev inputs. Means normalized by E_N, with sample SDs in parentheses. The disagreement term is nonnegative; the directional term need not be.

| Campaign | Support | Disagreement | Directional cross term | Final error difference |
|---|---|---|---|---|
| ID2 | Native | 0.088504 (0.003039) | −0.034664 (0.004404) | 0.053840 (0.002857) |
| ID2 | Q_pos | 0.037939 (0.003012) | −0.044936 (0.005606) | −0.006998 (0.003336) |
| XR1 | Native | 0.025893 (0.001274) | −0.024657 (0.002432) | 0.001236 (0.001662) |
| XR1 | Q_pos | 0.025478 (0.001227) | −0.029207 (0.002338) | −0.003729 (0.001513) |
| CMR1 | Native | 0.013716 (0.001996) | −0.012329 (0.002098) | 0.001387 (0.001033) |
| CMR1 | Q_pos | 0.013997 (0.002160) | −0.015891 (0.002359) | −0.001894 (0.001227) |

These identities describe error geometry. They do not assign causal roles to either term or identify an independent mediator.

## E ID2 short-horizon cross-continuation

### E.1 Fixed local intervention

Replicates 01–08 were selected by index before continuation, without screening target outcomes or diagnostic quality. Each historical terminal checkpoint seeded two branches: A, Native start → pure Q_pos; B, Native start → 0.9 Q_pos + 0.1 Native; C, Q_pos start → pure Q_pos; D, Q_pos start → the same mixture. All 32 branches ran exactly 1000 updates, with audit scoring at 0/100/300/1000 and donor-aligned dev diagnostics at 0/1000. All times are reported; the audit did not select a checkpoint.

Before any smoke updates, historical train/dev/audit banks (4096/256/512 rows) were reconstructed with their original frozen roles and matched donor, permutation, hidden, and Teacher-label hashes. No substitute audit was generated. Step-0 scoring was reproducible, and the response/decomposition baseline exactly reproduced each selected replicate's first-stage diagnostic.

AdamW state was reset to empty at every branch. Other optimizer settings were retained, but a fixed 50-step warmup to 3e-4 and cosine decay to 3e-5 replaced the old schedule. The historical terminal learning rate was actually 1e-4, not zero. Each update used 32 Student forward samples and four microbatches of eight. Pure branches used two independent Q_pos batches of 16 with weights 0.5/0.5; mixed branches used Q_pos and Native batches of 16 with weights 0.9/0.1. This explicitly estimates the complete-bank weighted MSE objective, without row replacement or a claim of a marginal-preserving mixed minibatch. All index streams were frozen before training. Updates and forward samples were matched, but gradient variance, supervision information, and number of Q_pos samples were not identical.

Actual training totals were 32,000 updates, 1,024,000 forward samples, and 128,000 Student microbatch forwards, excluding the separate eight-update smoke. Thirty-eight archived tests passed. Continuation artifacts remained separate from historical checkpoints; no formal target score was computed.

## E.2 All scheduled audit results

**Table E1:** Audit trajectories and full-training-support objective. NMSE is mean ± sample SD across eight pairs. SSE columns are replicate means in millions. Each audit column has its own frozen denominator shared across branches. Objective values are MSE averaged over each branch's complete original training supports with its specified weights, not normalized errors. Result identity is recorded in Appendix A.

| Branch | Step | Native NMSE | Q_pos NMSE | Native SSE(M) | Q_pos SSE(M) | Training-objective MSE |
|---|---|---|---|---|---|---|
| A | 0 | 0.103499 ± 0.002394 | 0.104059 ± 0.003288 | 2.724014 | 2.617068 | 0.051088 |
| A | 100 | 0.108525 ± 0.002591 | 0.100812 ± 0.002938 | 2.856303 | 2.535395 | 0.049271 |
| A | 300 | 0.109338 ± 0.002811 | 0.099260 ± 0.003013 | 2.877698 | 2.496358 | 0.048336 |
| A | 1000 | 0.108375 ± 0.002805 | 0.096698 ± 0.002967 | 2.852365 | 2.431935 | 0.046923 |
| B | 0 | 0.103499 ± 0.002394 | 0.104059 ± 0.003288 | 2.724014 | 2.617068 | 0.051165 |
| B | 100 | 0.106727 ± 0.002534 | 0.101779 ± 0.003012 | 2.808965 | 2.559708 | 0.050199 |
| B | 300 | 0.106225 ± 0.002481 | 0.100151 ± 0.002963 | 2.795771 | 2.518785 | 0.049335 |
| B | 1000 | 0.103940 ± 0.002510 | 0.097238 ± 0.002962 | 2.735618 | 2.445517 | 0.047818 |
| C | 0 | 0.156142 ± 0.000713 | 0.094856 ± 0.000616 | 4.109563 | 2.385630 | 0.045492 |
| C | 100 | 0.156805 ± 0.000648 | 0.095882 ± 0.000623 | 4.127012 | 2.411422 | 0.046012 |
| C | 300 | 0.156297 ± 0.000559 | 0.095196 ± 0.000586 | 4.113622 | 2.394168 | 0.045675 |
| C | 1000 | 0.154399 ± 0.000403 | 0.093287 ± 0.000486 | 4.063672 | 2.346157 | 0.044720 |
| D | 0 | 0.156142 ± 0.000713 | 0.094856 ± 0.000616 | 4.109563 | 2.385630 | 0.048992 |
| D | 100 | 0.150702 ± 0.000694 | 0.096789 ± 0.000623 | 3.966391 | 2.434248 | 0.049606 |
| D | 300 | 0.149056 ± 0.000867 | 0.096174 ± 0.000653 | 3.923062 | 2.418765 | 0.049260 |
| D | 1000 | 0.146111 ± 0.000410 | 0.093912 ± 0.000516 | 3.845548 | 2.361879 | 0.048093 |

**Table E2:** End-minus-baseline audit changes. Mean ± sample SD of paired changes, not differences of unpaired error bars. Positive means worse. Both supports improve in all eight C and D runs; Native audit worsens in all eight A and B runs.

| Branch | Native change | Q_pos change |
|---|---|---|
| A | +0.004877 ± 0.000762 | −0.007361 ± 0.000700 |
| B | +0.000441 ± 0.000275 | −0.006821 ± 0.000703 |
| C | −0.001744 ± 0.000325 | −0.001569 ± 0.000145 |
| D | −0.010031 ± 0.000415 | −0.000944 ± 0.000115 |

Compared with same-start pure continuation, B−A and D−C Native-column differences are −0.004436 and −0.008288 at step 1000 (all eight pairs negative), with Q_pos-column costs +0.000540 and +0.000625 (all eight positive). No registered preservation tolerance applies to

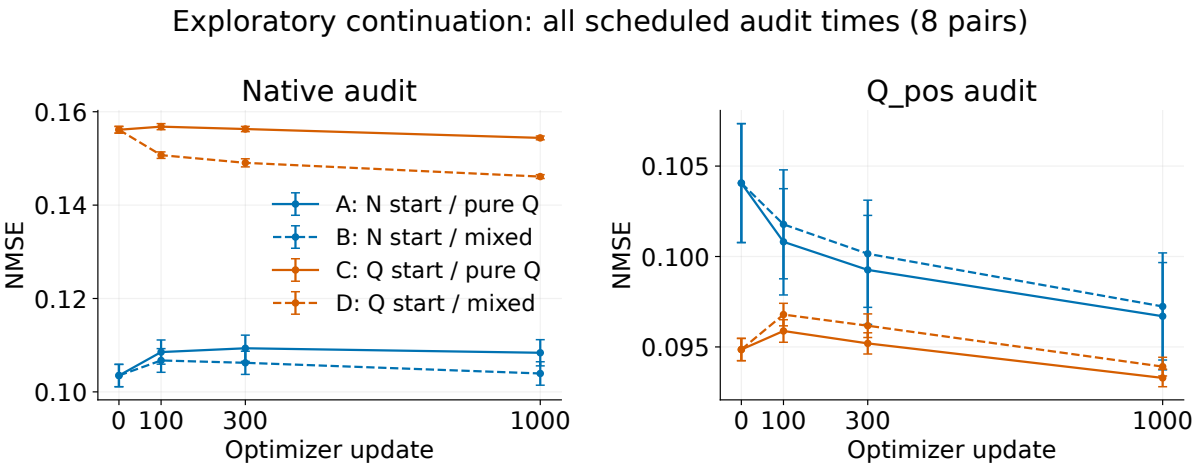



**Figure E1:** Fixed-horizon continuation. All four branches and all preselected times are shown. Points and bars are mean ± one sample SD across eight replicates; connecting segments guide the eye and do not assert intermediate measurements. Blue denotes the Native start, orange the Q_pos start; solid denotes pure Q_pos, dashed the mixed objective. The two panels have different audit denominators and are interpreted within a panel. Data come exclusively from the persisted continuation summary.

these costs. Native-column starting-point gaps remain +0.046023 for C−A and +0.042171 for D−B, versus +0.052643 initially. Pure Q_pos continuation does not remove the historical Q Student's Native deficit; the mixture changes it more but does not restore the historical Native baseline. Both C and D improve both supports relative to their baselines, so a local relative tradeoff is not proof that improvement on Native must worsen initial Q_pos performance or that a hard capacity limit has been reached.

### E.3 Functional changes and nonmediation boundary

**Table E3:** Step-0 → step-1000 donor-aligned dev response reproduction. These are the fixed eight-replicate subset, not the 48-pair means in Table D2.

| Branch | Relative response error | Projection gain | Cosine |
|---|---|---|---|
| A | 0.257428 → 0.261398 | 0.745224 → 0.734690 | 0.861658 → 0.859333 |
| B | 0.257428 → 0.241351 | 0.745224 → 0.759731 | 0.861658 → 0.870925 |
| C | 0.470617 → 0.468417 | 0.527092 → 0.530045 | 0.727592 → 0.729099 |
| D | 0.470617 → 0.455699 | 0.527092 → 0.540927 | 0.727592 → 0.737780 |

The mixed Native-start branch B improves response error, gain, and cosine in all eight replicates while slightly worsening Native audit in all eight. C/D improvements are accompanied by final-error improvements but do not demonstrate a causal mediator. On Native dev inputs, all four branches reduce the attention, fixed-state MLP, and propagation squared norms on average, while weakening negative attention–MLP compensation. For A, attention and MLP squared changes −0.010887/−0.022761 coexist with a +0.037910 attention–MLP cross-term change and a +0.004867 full-error change (normalized by dev E_N). For D, attention squared changes by −0.014076, the same cross term by +0.010894, and full error by −0.010026. Audit and dev are independent supports, so their similar directions are not identical measurements. All terms and cross terms remain in the archived summaries.

Extra optimization, support-specific supervision, and starting-point effects coexist within this short local intervention; their independent contributions remain unresolved.

# F Elementary theoretical clarification

Consider a scalar query $x$ and a preceding scalar $z$. Native's four-row bank is two copies each of $(z, x) = (-1, -1)$ and $(1, 1)$. Independently permuting the history column can produce all four sign combinations once while preserving both finite column multisets. Let $T(z, x) = x$, independent of history, and restrict the Student to $S_w(z, x) = wz$. The complete-bank risks are

$$L_N(w) = (w - 1)^2, \qquad L_Q(w) = w^2 + 1.$$

Their optima are $w = 1$ and $w = 0$. On Native evaluation the respective squared risks are 0 and 1, with correct Teacher relabeling and a history-insensitive Teacher.

The example establishes a logical separation between Teacher history sensitivity and support dependence in restricted learning. Its Student can use only history, a structural restriction unlike the actual Transformer Student; it therefore does not explain ID2's mechanism. If both Teacher and Student are strictly pointwise at each position and loss is additive over positions, the complete-bank empirical objective is invariant under Q_pos; a finite optimizer can still depend on minibatch grouping. General theoretical possibilities and the project's measured Transformer behavior should not be conflated.